

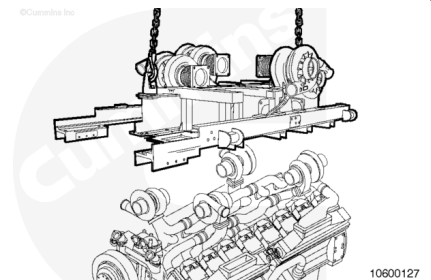
## Trainings ▾

### Install

#### with Mechanically Actuated Injector

##### **⚠ WARNING ⚠**

**This assembly weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this assembly.**



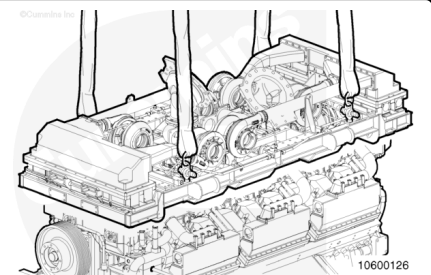
Connect the lifting straps to the two-stage frame lifting eyes.

Position and align the two stage frame on the three center support brackets on the engine.

Do **not** tighten the two-stage frame assembly capscrews until all capscrews have been in the supports and brackets.

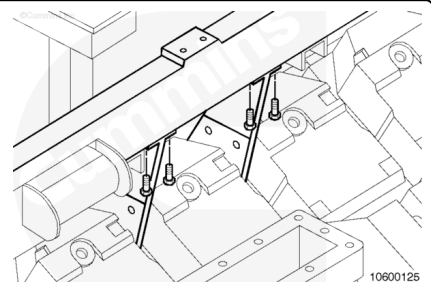
Align the two-stage frame assembly capscrew holes with the center support bracket capscrew holes.

Install the 10 capscrews connecting the two-stage frame to each of the center support brackets.



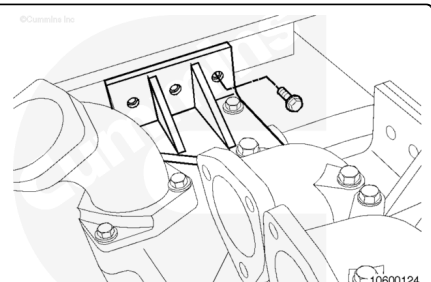
Align the two-stage frame assembly capscrew holes with the outboard support brackets.

Install the capscrews connecting the two-stage frame to each of the outboard support brackets.



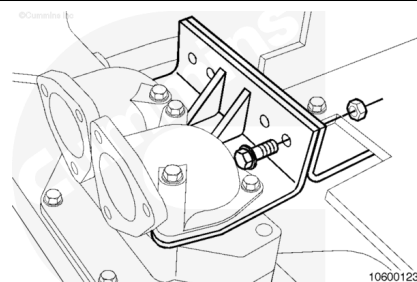
Align the two-stage frame assembly capscrew holes with the two low temperature thermostat housing bracket capscrew holes.

Install the capscrews connecting the two-stage frame to each of the two low temperature aftercooling thermostat housing brackets.

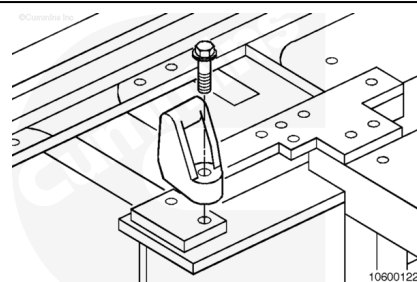


Tighten all of the two-stage frame assembly mounting capscrews.

**Torque Value:** 80 n·m [ 59 ft-lb ]



Remove the lifting eyes from the two-stage frame.

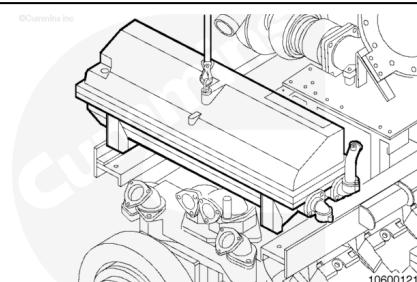


**⚠ WARNING ⚠**

**This assembly weighs 23 kg [50 lb] or more. To reduce the possibility injury, use a hoist or get assistance to remove this assembly.**

Install a 12 mm [0.47 in] lifting device in the top of the intercoolers.

Install the Intercoolers.

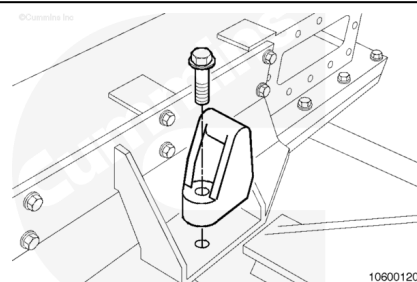


Remove the lifting eye from the intercoolers.

Install the lifting eye onto the two-stage frame assembly.

Tighten the lifting eye capscrew.

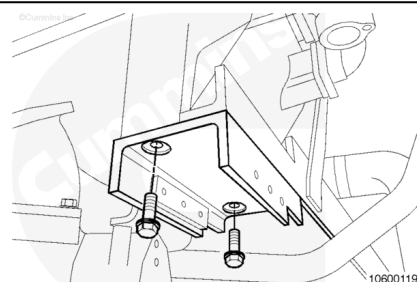
**Torque Value:** 195 n·m [ 144 ft-lb ]



Install the intercooler mounting capscrews.

Tighten the capscrews.

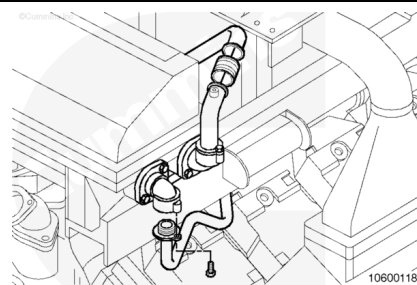
**Torque Value:** 280 n·m [ 207 ft-lb ]



Install the coolant supply and return piping onto the intercoolers.

Tighten the capscrews.

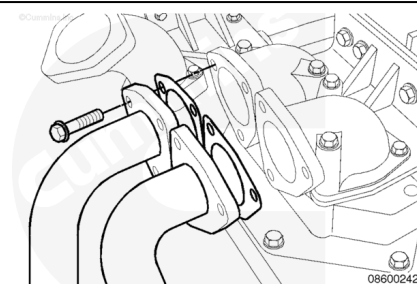
**Torque Value:** 45 n•m [ 33 ft-lb ]



Install the low temperature aftercooling thermostat housing coolant supply lines.

Tighten the capscrews.

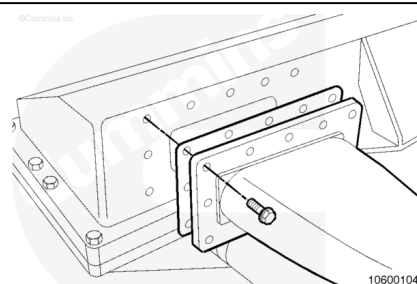
**Torque Value:** 54 n•m [ 40 ft-lb ]



Install the four intercooler intake crossovers.

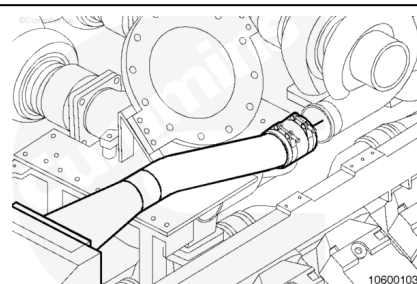
Tighten the capscrews.

**Torque Value:** 45 n•m [ 33 ft-lb ]

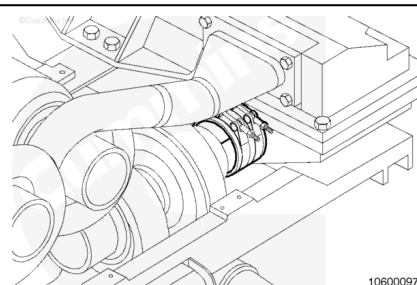


Install the hoses and clamps on the right-bank rear and left-bank front high-pressure turbochargers.

**Torque Value:** 9 n•m [ 80 in-lb ]



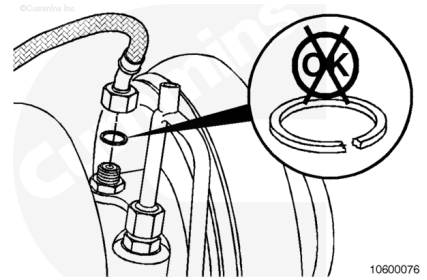
Install the intake hoses and clamps on each of the four high-pressure turbochargers.



Make sure the oil supply line o-ring is **not** damaged.

Install the oil supply lines on the four high-pressure turbochargers and tighten the fittings.

**Torque Value:** 54 n•m [ 40 ft-lb ]

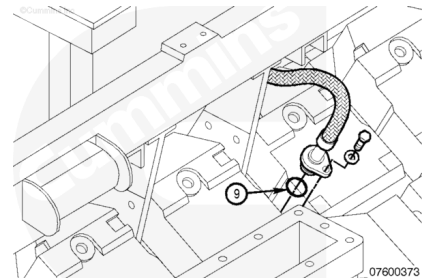


Install new o-ring seals on all of the turbocharger oil drain lines.

Install the oil drain lines into the cylinder block.

Tighten the oil drain line capscrews.

**Torque Value:** 45 n•m [ 33 ft-lb ]

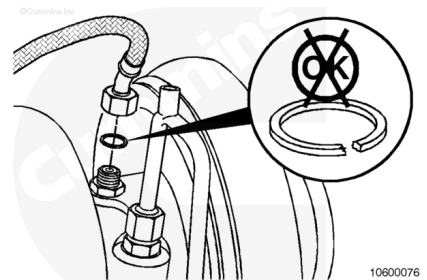


Make sure the oil supply line o-ring is **not** damaged.

Install the low-pressure turbocharger oil line to each low-pressure turbocharger.

Tighten the fittings.

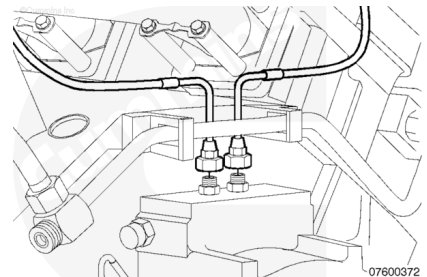
**Torque Value:** 54 n•m [ 40 ft-lb ]



Connect the two oil supply lines at the flywheel housing.

Tighten the fittings.

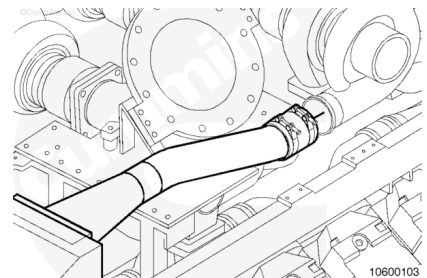
**Torque Value:** 54 n•m [ 40 ft-lb ]



Install the air crossover that connect the left bank front and right bank rear high-pressure turbocharger to the aftercoolers.

Install the clamps on the hose connections at the air crossover to the high-pressure turbochargers.

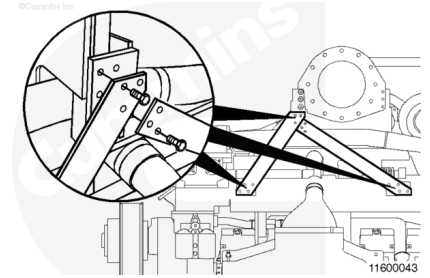
**Torque Value:** 9 n•m [ 80 in-lb ]



Install the air crossover supports.

Tighten the capscrews.

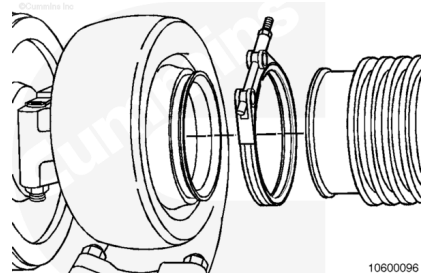
**Torque Value:** 45 n•m [ 33 ft-lb ]



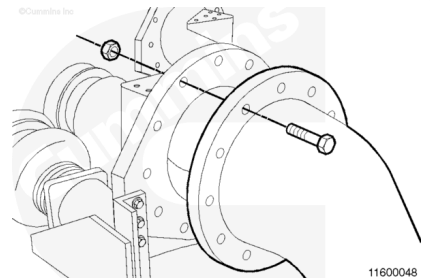
Install the V-band clamp and the exhaust bellows to the turbocharger exhaust outlet flanges.

Tighten the V-band clamp nut.

**Torque Value:** 9 n•m [ 80 in-lb ]



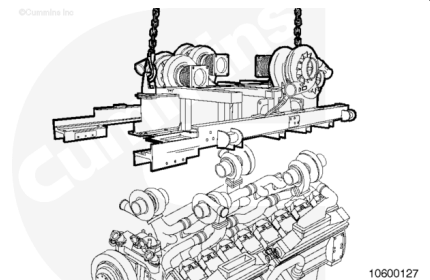
Connect the intake piping between the air filter(s) and the low-pressure turbochargers at the inlet side of the turbochargers.



## with Electronically Actuated Injector

### **⚠ WARNING ⚠**

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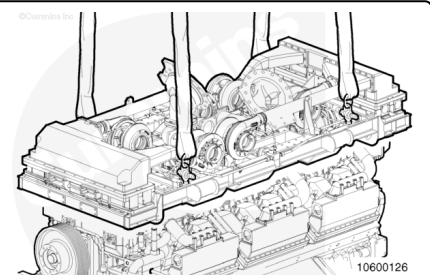
Connect the lifting straps to the two-stage frame lifting eyes.

Position and align the two stage frame on the three center support brackets on the engine.

Do **not** tighten the two-stage frame assembly capscrews until all capscrews have been in the supports and brackets.

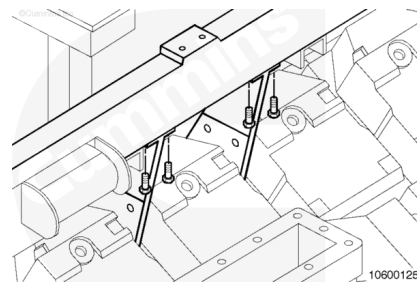
Align the two-stage frame assembly capscrew holes with the center support bracket capscrew holes.

Install the nine capscrews connecting the two-stage frame to each of the center support brackets.



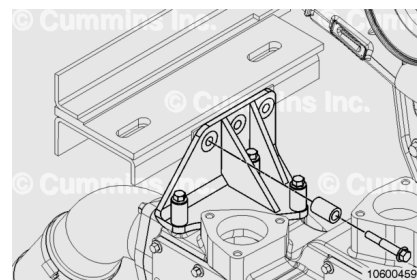
Align the two-stage frame assembly capscrew holes with the outboard support brackets.

Install the capscrews connecting the two-stage frame to each of the outboard support brackets.



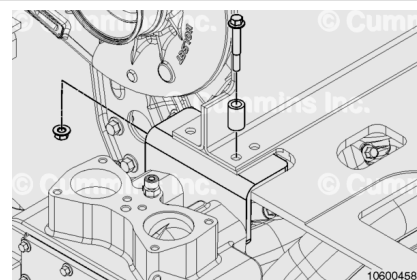
Align the two-stage frame assembly capscrew holes with the two low temperature thermostat housing bracket capscrew holes.

Install the capscrews connecting the two-stage frame to each of the two low temperature aftercooling thermostat housing brackets.



Tighten all of the two-stage frame assembly mounting capscrews.

**Torque Value:** 80 n•m [ 59 ft-lb ]



Connect the intake piping between the air filter(s) and the low-pressure turbochargers at the inlet side of the turbochargers.

